NON-DESTRUCTIVE TESTING OF GEARS

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Subject: Artificial Intelligence in Material Diagnostics

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OUTLINE

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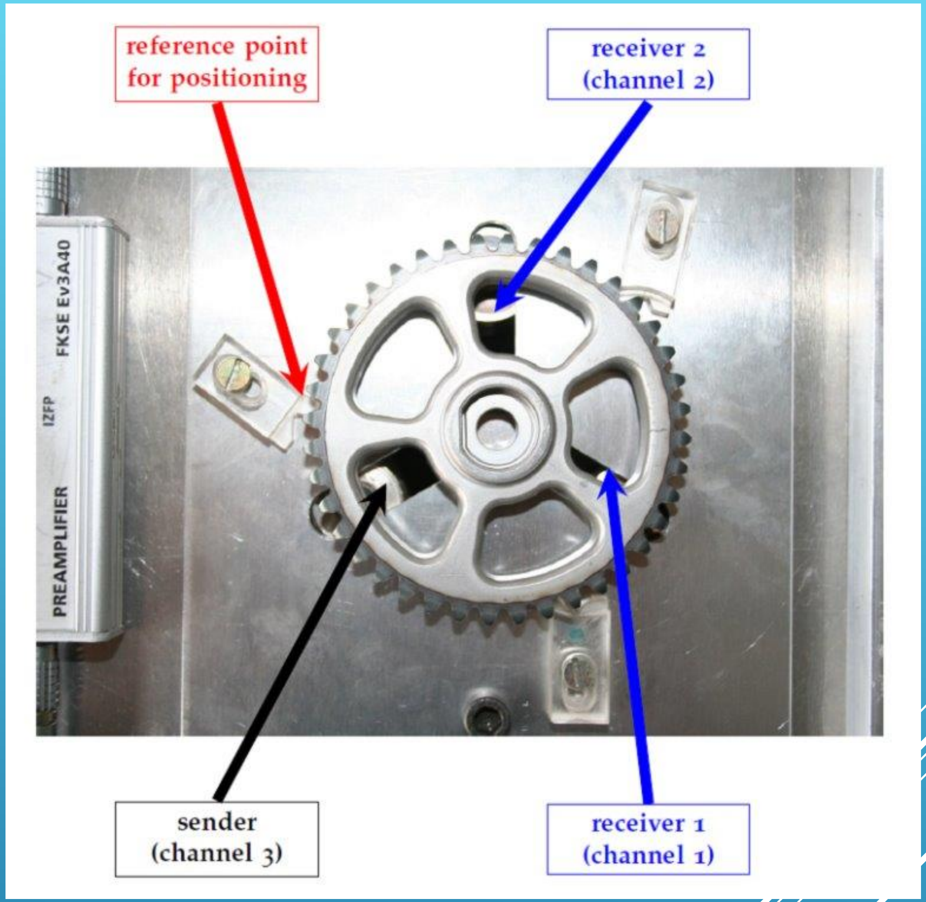


1. INTRODUCTION AND TASK

Material diagnostics is important to detect broken parts of a system like metal or glass

--> ensuring quality control and safety

--> non-destructive testing, so tested actually working materials stay intact

Task:

- Distinguish between good and bad gears using acoustic measurement signals
- Load the given signals into a dataframe, calculate STFT and plot spectrograms
- Compare results and detect the broken gear

2. DATA

- one dataset of two gears

- For each gear four ultrasonic signals taken from two channels

- Gears are named Z01 and Z05

- One gear is broken and the other one is alright

3. PREPROCESSING

- We use os.path.join for reading the data
- We get all the data with .wav at the end
- In 'entry' we put the keys, according to the task
- After that we add it to our data list

```
pattern = os.path.join('data2025', 'Z*', 'Z*', 'Wav', '*.wav')
files = glob.glob(pattern)
print("Gefundene Dateien:", files)
```

```
data = []
```

```
entry = {
    'fn': fn,
    'sig': signal,
    'spec': parts[0],
    'mID': parts[1],
    'time': parts[2],
    'rID': parts[3][:3],
    'sID': parts[3].split('.')[0],
    'stft': None
}
data.append(entry)
```

4. STFT-CALCULATION

```
for idx, row in df.iterrows():
    fs = 1e6  # Anpassen an reale Abtastrate
    f, t, Zxx = stft(row['sig'],
                     fs=fs,
                     window='hann',
                     nperseg=256,     # Fensterlänge
                     noverlap=128)    # 50% Overlap

    df.at[idx, 'stft'] = (f, t, np.abs(Zxx))
```

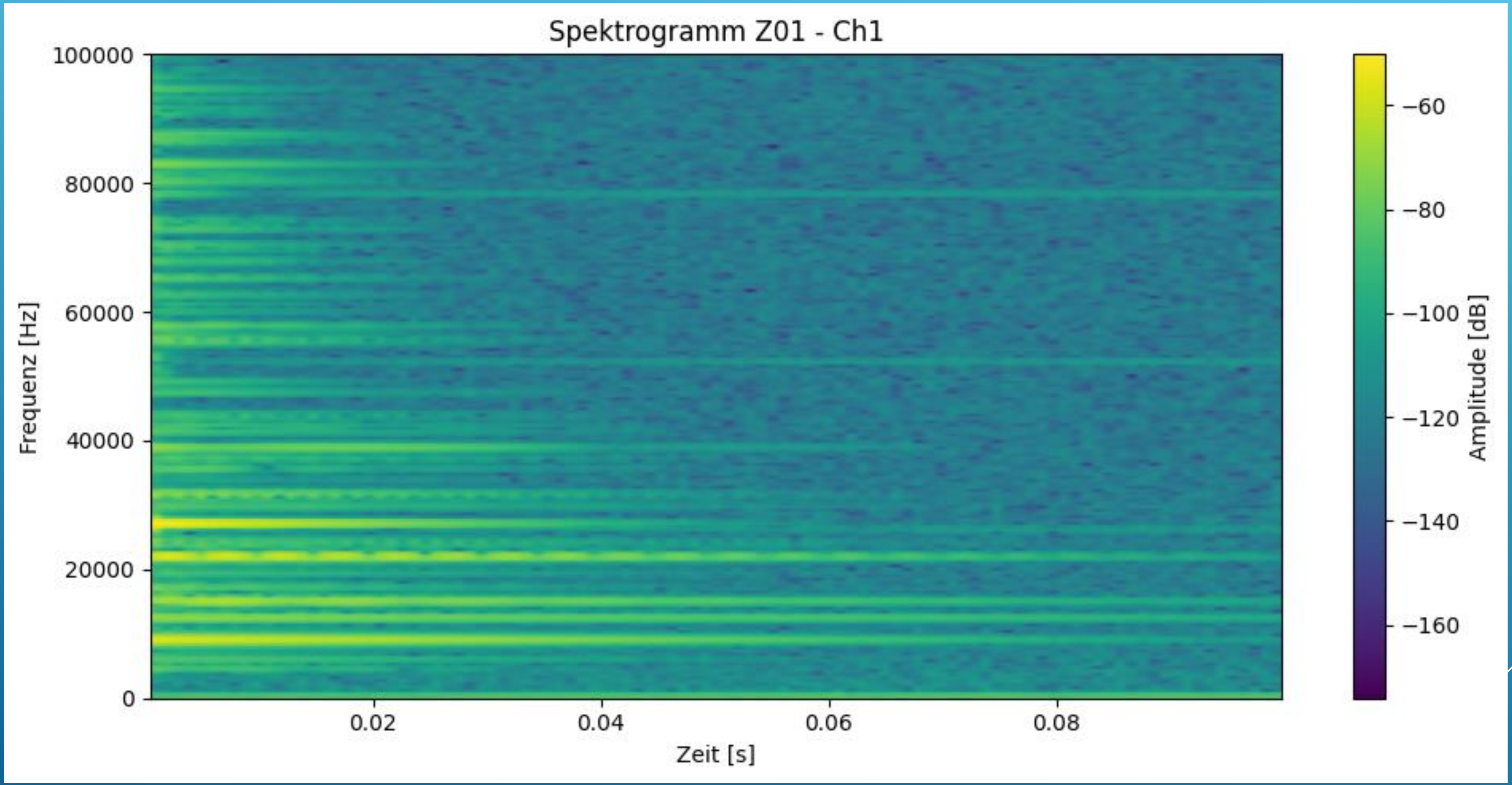
- Set sampling rate

- `f`: Array with frequencies (in Hz) for the STFT.
- `t`: Array with times we used for calculating STFT
- `Zxx`: Array with STFT-Coefficients (Amplitude and Phase).

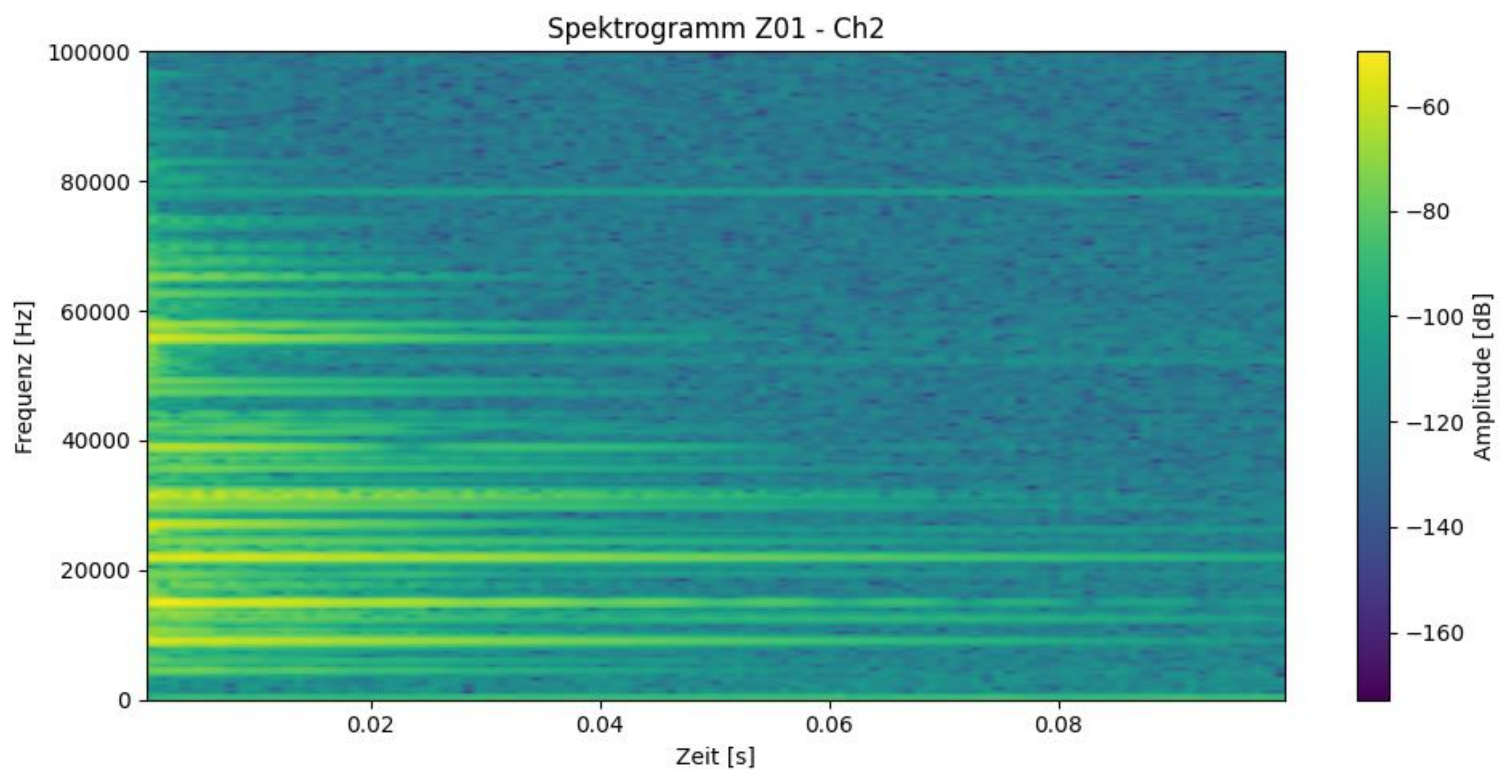
- Set window-options, for example length and overlap

- Save the result of the STFT as a triple (frequency, time, spectrum)

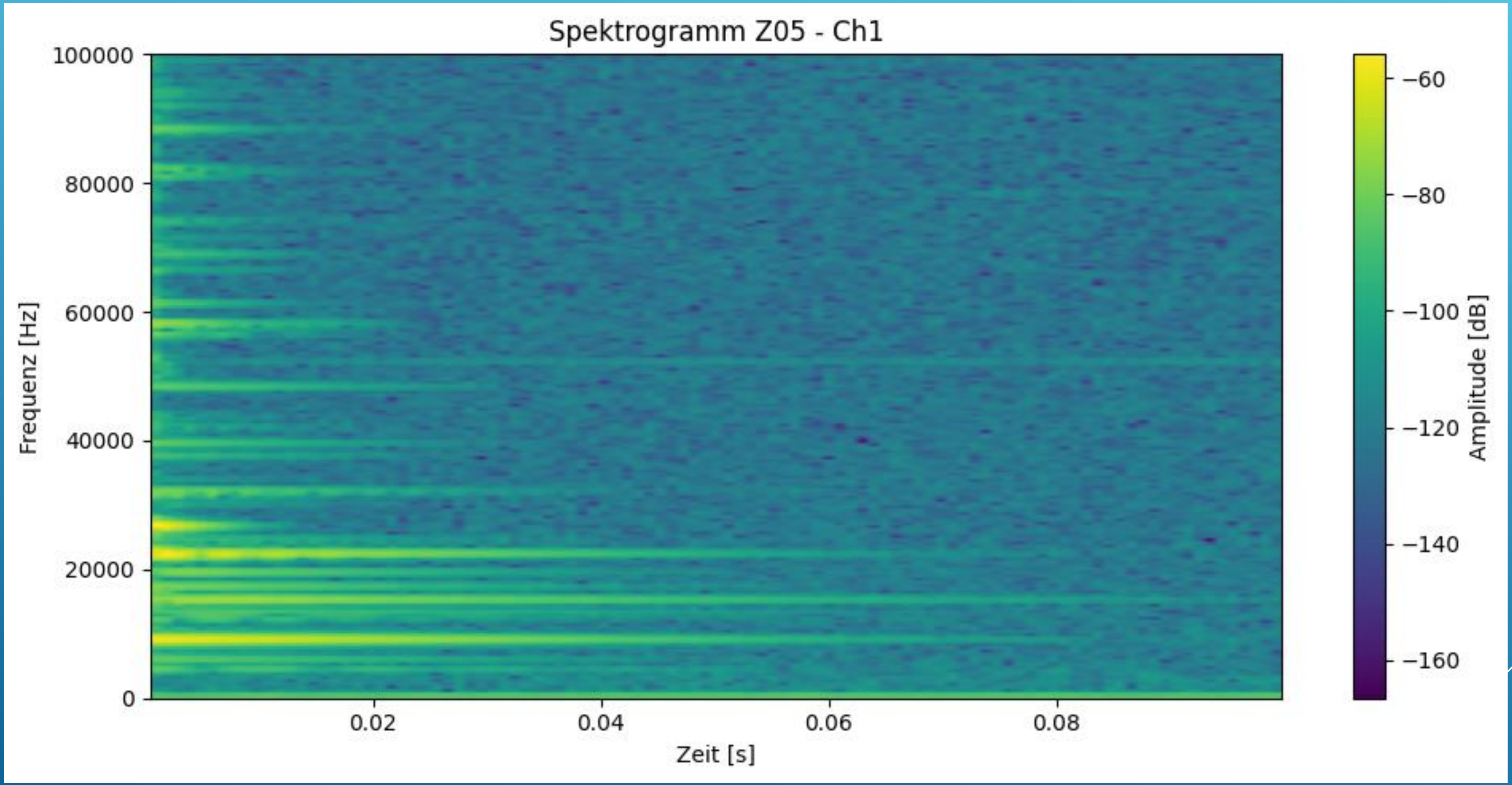
5. <u>RESULTS OF THE STFT-ANALYSIS</u> Z01 CH1



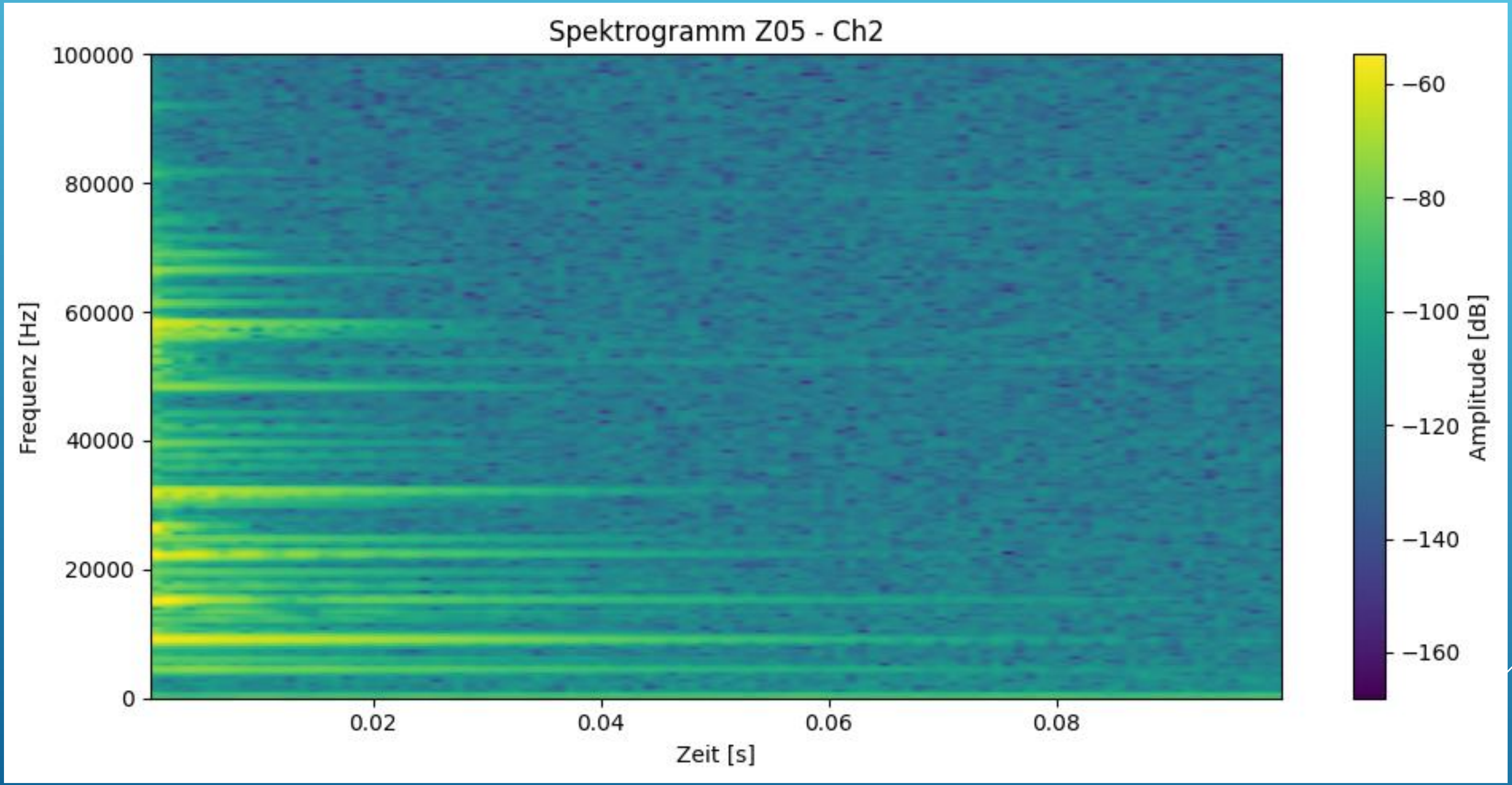
5. RESULTS OF THE STFT-ANALYSIS Z01 CH2



5. RESULTS OF THE STFT-ANALYSIS Z05 CH1



5. RESULTS OF THE STFT-ANALYSIS Z05 CH2



6. COMPARISON

- Frequencies in Z01 could be heard for a relatively long time
- In Z01 there are many noticeable measurements over 20 kHz
- Frequencies in Z05 are mostly shorter than in Z01
- In Z05 there are barely any good hearable (visible) frequencies above 20 kHz

7. CONCLUSION

- Z01 has higher and longer frequencies than Z05

- Those differences are hearable und visible

- We compared that situation with wine glasses:

  intact glass: longer and higher sounds

  broken glass: shorter and mostly deeper sounds

7. CONCLUSION

- That case transmitted to the gears we think that:
- Z01 is the intact gear
- Z05 is the broken gear

THANK YOU FOR YOUR ATTENTION